

Pulkit Kumar Bathwal



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INTRODUCTION

ECM student with strong foundation in algorithms and software development. Experienced in Python, C/C++, and Java with projects in machine learning, embedded systems, and predictive modeling. Active in competitive programming (300+ problems solved on LeetCode)

EDUCATION

Vellore Institue of Technology, Chennai

B.Tech Electronics and Computer

June 2027

Current GPA: 8.76/10.0

Senior Secondary

Delhi Public School, Raipur

June 2020 - March 2022

COURSEWORKS

Courses: Object-Oriented Programming, Data Structures & Algorithms, Embedded Systems, Discrete Math, Linear Algebra, Calculus, Physics, Probability & Statistics, Design and Analysis of Algorithm, Theory of Computation, Computer Network, Database Management Systems

SKILLS

Programming Languages: C/C++, Python, Java

Tools: Git/GitHub, VS Code, Kaggle, Embedded Systems, Microcontroller Programming, Antigravity

Machine Learning : Machine Learning (Advanced), NumPy, Pandas, Matplotlib, Data Analysis, Predictive Modeling

PROJECTS

RAG based PDF chatbot | *Python, Machine Learning*

July 2026

- Developed a full-stack Retrieval-Augmented Generation (RAG) application using FastAPI and Streamlit, leveraging PostgreSQL (pgvector) to index and retrieve document chunks.
- Integrated Gemini API via the new google-genai SDK to generate vector embeddings and deliver precise, context-grounded answers with automatic page citations.

Multi-Agent Research Assistant | *Python, Machine Learning*

July 2026

- Orchestrated via a LangGraph state graph, the system executes concurrent web-search (Tavily) and vector-database (pgvector) RAG pipelines, bound by an SQLite-persisted interrupt() state machine for asynchronous human approval gates..
- Server-Sent Events (SSE) stream task-level updates using an async astream() generator, tracking real-time API pricing and aggregating input/output token cost fields via Pydantic state reducers..

Developer Portfolio Website | *React.js, JavaScript, HTML, CSS*

Feb 2026

- Designed and developed a personal portfolio website using React.
- Showcased projects, technical skills, and coding profiles with responsive UI.
- Implemented reusable components and modern frontend design practices.

CERTIFICATION

Machine Learning Specialization (Coursera, Andrew Ng) — Completed

December, 2025

- Mastered supervised learning techniques, including Linear Regression, Logistic Regression, and advanced Classification models.
- Developed a strong theoretical foundation and practical skills in implementing modern ML workflows.